PRACTITIONER KNOWLEDGE REPOSITORY

MODULE 4

Funding Valuation Adjustment (FVA) & Margin Valuation Adjustment (MVA)

The Hidden Tax on Every Derivative You Trade

Sarthaka Nanda | with Anthropic Claude

Practitioner Knowledge Repository · 2025 | CC BY-NC 4.0

TABLE OF CONTENTS

1. Layman Analogy — The Funding Cost Hiding Below the Waterline 4
The Mortgage Analogy for FVA 4
The Pension Fund Analogy for MVA 4
2. The Problem FVA Solves: When Risk-Neutral Pricing Breaks Down 5
The Funding Gap 5
3. FVA & MVA Defined: A Side-by-Side Introduction 6
Quick Intuition: The Core Formulas 6
Where FVA & MVA Sit in the XVA Hierarchy 6
4. FVA: Formal Framework & Formula Derivation 8
4.1 The Piterbarg Framework (2010) 8
4.2 FCA and FBA: The Two Components of FVA 8
4.3 Practical Computation 8
5. The Hull-White Controversy: Should FVA Exist? 10
The Monetisation Problem 10
JP Morgan's Historic FVA Recognition (January 2014) 10
6. DVA vs FVA: The Double-Count Problem 11
The Overlap 11
The Burgard-Kjaer Resolution 11
7. Worked Numerical Example: FVA on an Uncollateralised IRS 12
Trade Setup 12
Step 1: EPE and ENE Profile 12
Step 2: Net FVA and Key Insight 13
8. MVA: Formal Framework & Formula Derivation 14
8.1 Origins: The UMR Revolution 14
8.2 The ISDA SIMM Model 14
8.3 MVA Formula and Intuition 14
9. Worked Numerical Example: MVA Under ISDA SIMM 16
Trade Setup (same EUR 100M 5Y IRS as Section 7) 16
Step 1: SIMM IM at Inception 16
Step 2: IM Amortisation Profile & MVA Calculation 16
Step 3: Total XVA for This Trade 16
10. Regulatory Treatment: FRTB-CVA, SA-CCR & UMR Interactions 18
10.1 FVA and Regulatory Capital 18
10.2 MVA and the UMR Regulatory Framework 18

10.3 FRTB-CVA Interaction 18
10.4 The Cleared vs Bilateral Choice 18
11. Crisis Case Studies 20
11.1 Lehman Brothers Collapse (September 2008) 20
11.2 The 2008–2009 Funding Crisis: Systemic FVA Implications 20
11.3 JP Morgan FVA Charge (January 2014) 20
11.4 UMR Phase 6 (September 2022): The MVA Inflection Point 20
12. Advanced Topics 22
12.1 FVA Asymmetry: Wrong-Way Risk 22
12.2 Bilateral vs Unilateral FVA 22
12.3 ColVA — Preview of Module 6 22
12.4 The XVA Optimisation Problem 22
13. References 24
Regulatory & Official Documents 24
Academic Papers 24
Accounting & Valuation Standards 24
Market & Industry Sources 24
14. Credits & Methodology Note 25
Author 25
AI Collaboration 25
Series 25
Licence 25

1. Layman Analogy — The Funding Cost Hiding Below the Waterline

Imagine you are a bank. A corporate client asks you to enter a 5-year interest rate swap — they pay fixed, you pay floating. In a textbook world, you price this at fair value using risk-neutral discounting: expected cash flows discounted at the risk-free rate. Done.

But here is what the textbook omits: **you need money to fund this position.** If the swap moves in your favour you have a receivable — but you cannot borrow against it at the risk-free rate. You borrow at your own credit spread, which may be 50, 80, or 150 basis points above overnight rates. Conversely, if the swap moves against you, your counterparty will not post collateral unless a CSA (Credit Support Annex) is in place. You are funding that exposure out of your own treasury.

Think of an iceberg. What markets see — and what traditional pricing models capture — is the tip: the mark-to-market value. Below the waterline sit the real economic costs that erode P&L:

- **FVA (Funding Valuation Adjustment):** The cost of funding uncollateralised derivative exposures over the life of the trade.
- **MVA (Margin Valuation Adjustment):** The ongoing cost of posting initial margin — cash or securities locked with a custodian earning below-market returns.
- **CVA/DVA** (Module 3): Credit valuation adjustments for counterparty default risk.

Together, these adjustments — collectively called XVA — represent the true all-in cost of a derivative. A bank that ignores FVA and MVA is not mispricing its book; it is subsidising its clients from shareholder capital without knowing it.

The Mortgage Analogy for FVA

Suppose you lend a friend £100 for a year. He pays you back with interest. But you borrowed that £100 from your bank at 4% to fund the loan. Even if your friend repays exactly as agreed, you still incurred a real cost of borrowing. FVA is precisely this: the cost a dealer bank incurs to fund its derivatives book when the market does not automatically return collateral to offset those positions.

The Pension Fund Analogy for MVA

Under post-2008 regulation (EMIR, Dodd-Frank, UMR), most non-cleared derivatives now require both parties to post Initial Margin (IM) — a buffer against future potential exposure. For a pension fund or corporate, this IM must be funded: either by selling assets or borrowing cash. The return earned on posted IM (typically overnight rate) is almost always below the fund's hurdle rate. That drag over the life of the trade is MVA.

2. The Problem FVA Solves: When Risk-Neutral Pricing Breaks Down

The classical derivative pricing framework — pioneered by Black-Scholes (1973) and extended through risk-neutral measure theory — rests on a critical assumption: perfect capital markets with costless, unlimited funding at the risk-free rate. In this world, a replicating portfolio can always be constructed without friction. Funding is free. Collateral is irrelevant. The no-arbitrage price is unique.

The 2008 global financial crisis exposed every one of these assumptions as fiction:

Textbook World (Pre-2008)	Real World (Post-2008)
• Funding available at risk-free rate • Unlimited borrowing capacity • Collateral symmetry assumed • Single discount curve: LIBOR ≈ OIS • Price is unique and model-independent	• Banks fund at OIS + credit spread (30–150 bps) • Funding capacity is constrained • Asymmetric CSA terms are common • Multi-curve: OIS ≠ LIBOR ≠ funding rate • Price depends on who funds the trade

The multi-curve framework emerged first — separating OIS discounting from LIBOR projection curves. But it still did not capture the dealer's own funding cost. FVA emerged as the formal mechanism to bring that cost into pricing.

The Funding Gap

Consider a dealer bank with a portfolio of uncollateralised OTC derivatives. At any point in time, some trades are in-the-money (assets) and some are out-of-the-money (liabilities). The net positive market value represents an economic receivable — but the bank cannot immediately realise it. To fund its operations in the interim, the bank borrows in the wholesale market at a spread above OIS. This spread — the

funding spread or *treasury spread* — is typically 30–150 bps depending on the bank's credit quality and market conditions.

Market Condition	Bank Funding Spread (bps over OIS)	FVA Impact: EUR 100M 10Y IRS
Normal (pre-stress)	30–50 bps	≈€150K–250K
Moderate stress	80–120 bps	≈€400K–600K
Crisis (2008-style)	200–400 bps	≈€1M–2M

Illustrative estimates for a plain vanilla EUR 100M notional 10-year IRS (uncollateralised).

3. FVA & MVA Defined: A Side-by-Side Introduction

FVA and MVA are related but distinct: both represent the cost of capital tied up in a derivative transaction, but they arise from different sources — unsecured funding gaps versus regulatory margin requirements.

FVA — Funding Valuation Adjustment	MVA — Margin Valuation Adjustment
• Arises from: uncollateralised derivative exposures • Driver: dealer funding spread above OIS (30–150 bps) • Applies to: bilateral OTC trades without full CSA • Direction: FCA (cost) when ITM; FBA (benefit) when OTM • Post-2008 catalyst: multi-curve divergence & funding stress • Key debate: Hull-White (2012) — "FVA is a mistake" • Typical magnitude: 10–80 bps of notional	• Arises from: mandatory Initial Margin (IM) under UMR • Driver: opportunity cost of segregated posted collateral • Applies to: bilateral & cleared trades (post Phase 6, 2022) • Direction: always negative — IM posting is always a cost • Post-2015 catalyst: BCBS-IOSCO UMR phase-in 2016–2022 • Computed via: ISDA SIMM methodology (v2.6) • Typical magnitude: 5–40 bps of notional

Quick Intuition: The Core Formulas

FVA — Continuous form:

$$FVA = \int_0^T s^o(t) \cdot [E[V^-(t)] - E[V^+(t)]] \cdot B(0,t)\, dt$$

where: $s^o(t)$ = funding spread at time t; $V^+(t) = \max(V(t),0)$ = positive exposure (FCA cost); $V^-(t) = \min(V(t),0)$ = negative exposure (FBA benefit); B(0,t) = OIS discount factor

FVA — Discrete (simulation) form:

$$FVA \approx -\Sigma^I_{t0}\, s^o(\Delta t) \cdot EPE(t^I) \cdot B(0,t^I) \cdot \Delta t + \Sigma^I_{t0}\, s^o(\Delta t) \cdot ENE(t^I) \cdot B(0,t^I) \cdot \Delta t$$

$EPE(t^I)$ = Expected Positive Exposure at time step i = $E[\max(V(t^I),0)]$ (amount the dealer must fund); $ENE(t^I)$ = Expected Negative Exposure = $E[\min(V(t^I),0)]$ (funding received from counterparty); s^o = bank funding rate minus risk-free/collateral rate; Δt = time step interval

MVA — Continuous form:

$$MVA = -\int_0^T s_m(t) \cdot E[IM(t)] \cdot B(0,t)\, dt$$

where: $s_m(t)$ = spread between dealer funding cost and IM remuneration rate (typically OIS); E[IM(t)] = expected SIMM or CCP initial margin requirement at time t; B(0,t) = OIS discount factor

The key insight is that both are path-dependent: they depend on the expected future profile of exposures or margin requirements, integrated over the entire life of the trade. This makes them computationally intensive — typically requiring Monte Carlo simulation of the underlying risk factors.

Where FVA & MVA Sit in the XVA Hierarchy

Adjustment	What It Prices	Who Bears Cost	Module
CVA	Counterparty default risk (expected loss)	Dealer (buyer of protection)	Module 3
DVA	Own default benefit to counterparty	Counterparty	Module 3
FVA	Funding cost of unsecured exposure	Dealer treasury	Module 4 ✓
MVA	Cost of posting initial margin	Both parties (UMR)	Module 4 ✓
KVA	Return on regulatory capital consumed	Shareholders	Module 5
ColVA	Cheapest-to-deliver collateral optionality	Collateral poster	Module 6

4. FVA: Formal Framework & Formula Derivation

4.1 The Piterbarg Framework (2010)

The most influential early formalisation of funding costs in derivatives was Vladimir Piterbarg's 2010 paper "Funding Beyond Discounting: Collateral Agreements and Derivatives Pricing." Piterbarg showed that when collateral and funding rates differ, the risk-free discount rate must be replaced by a funding-adjusted rate.

Consider a derivative with value *V(t)* and a perfect CSA (full cash collateral). The collateral account earns the overnight rate *rc.* The replication argument gives the following PDE:

$$\partial V/\partial t + \tfrac{1}{2}\sigma^2 S^2 (\partial^2 V/\partial S^2) + r^o \cdot S(\partial V/\partial S) - rc \cdot V = 0$$

Piterbarg PDE: r^O = funding rate of the replicating/hedging portfolio; rc = collateral (CSA) rate; σ = volatility of underlying S

Key result: when collateral is posted at the OIS rate $rc = r^{ol}_s$, the dealer discounts at OIS regardless of its own credit. But when the position is *uncollateralised,* the dealer discounts at its own funding rate $r^O = r^{ol}_s + s^O$, where s^O is the funding spread above OIS.

4.2 FCA and FBA: The Two Components of FVA

Burgard and Kjaer (2011) extended Piterbarg's framework to derive FVA as the sum of two components with opposite signs:

Component	Economic Meaning	Sign	Discrete Formula
FCA (Funding Cost Adjustment)	Cost of funding positive exposures (EPE). When the trade is in-the-money, the dealer must borrow to fund the receivable at spread s^O above OIS.	Negative (cost)	$FCA = -\Sigma\, s^O(t^I) \cdot EPE(t^I) \cdot B(0,t^I) \cdot \Delta t$
FBA (Funding Benefit Adjustment)	Benefit from receiving funding when trade is out-of-the-money (ENE). Counterparty effectively lends to dealer; dealer saves s^O on that amount.	Positive (benefit)	$FBA = +\Sigma\, s^O(t^I) \cdot \lvert ENE(t^I)\rvert \cdot B(0,t^I) \cdot \Delta t$

Combined:

$$FVA = FCA + FBA = \Sigma^I\, s^o(t^I) \cdot [\lvert ENE(t^I)\rvert - EPE(t^I)] \cdot B(0,t^I) \cdot \Delta t$$

$EPE(t^I) = E[max(V(t^I),0)]$ = Expected Positive Exposure; $ENE(t^I) = E[min(V(t^I),0)]$ = Expected Negative Exposure; $B(0,t^I)$ = OIS discount factor; Δt = simulation time step; s^O = bank unsecured funding rate minus OIS (the funding spread)

4.3 Practical Computation

In practice, FVA is computed via Monte Carlo simulation of the underlying market risk factors (interest rates, FX, credit spreads). At each simulation path and each time step, the portfolio value is computed, split into positive (EPE) and negative (ENE) components, multiplied by the funding spread, and discounted back to today. Key inputs:

- **Funding spread $s^o(t)$:** taken from the bank's internal transfer pricing curve, often proxied by CDS spreads or unsecured bond yields minus OIS.
- **EPE and ENE profiles:** the same Monte Carlo simulation used for CVA can be reused, reducing computational overhead significantly.
- **Netting and CSA terms:** FVA applies only to the uncollateralised portion of exposure. Trades under a full two-way daily-margined CSA have FVA ≈ 0.

5. The Hull-White Controversy: Should FVA Exist?

In 2012, John Hull and Alan White published "The FVA Debate" (Risk Magazine, July 2012), arguing that FVA is a mistake — that including it in derivative pricing destroys shareholder value and represents a fundamental error in financial reasoning. Their paper sparked the most intense academic debate in derivatives since Black-Scholes.

Hull-White (2012): FVA Should NOT Be Charged	Industry Response: FVA IS a Real Cost
• FVA is already captured in DVA (own-default benefit) • Modigliani-Miller: funding cost irrelevant to firm value • Charging FVA destroys positive NPV trades • Creates client-discriminatory, non-arbitrage-free pricing • Risk-neutral framework is internally complete • FVA is a shareholder transfer, not a real economic cost	• M-M assumes perfect markets — violated post-2008 • DVA cannot be monetised without the bank defaulting • Treasury charges desks real funding costs every day • Banks with high spreads genuinely cannot compete on price • Ignoring FVA creates systematic hidden P&L drag • Andersen-Duffie-Song (2019): FVA belongs in pricing

The Monetisation Problem

The crux of the debate is monetisation. Hull and White argued that DVA — the benefit a bank receives from the possibility of its own default — already captures the funding benefit (FBA). Practitioners responded: DVA cannot be realised without the bank actually defaulting. A bank cannot sell its own default risk in any liquid market. By contrast, FVA represents a real, daily cash cost: treasury literally charges the derivatives desk a funding spread every day. These are economically different things.

Andersen, Duffie and Song (2019) provided the most rigorous resolution: in a model where shareholder value maximisation is properly defined and funding markets are incomplete, FVA does belong in pricing — but only if correctly computed. The key is that FVA captures an opportunity cost to shareholders, not an accounting fiction. Most major dealers (JP Morgan, Goldman Sachs, Barclays) have since adopted FVA as a standard pricing component.

JP Morgan's Historic FVA Recognition (January 2014)

JP Morgan became the first major bank to publicly quantify and book an FVA loss: approximately USD 1.5 billion charged against Q4 2013 earnings. This was an accounting recognition that its uncollateralised derivatives portfolio had been implicitly subsidised by the firm's funding capacity, and that cost needed to be reserved. Goldman Sachs, Bank of America, and Morgan Stanley followed within months. The announcement validated the industry's practical approach and largely ended the academic debate at the practitioner level.

6. DVA vs FVA: The Double-Count Problem

A fundamental structural tension in XVA accounting is the potential overlap between DVA and FVA. Both involve the dealer's own creditworthiness, and naively including both can lead to double-counting.

The Overlap

DVA arises when the dealer is out-of-the-money (net liability to counterparty): the dealer may default before paying, so the counterparty discounts future payments at a rate reflecting dealer credit risk. The dealer books a gain.

FBA (the beneficial component of FVA) arises in the same out-of-the-money scenario: the counterparty is effectively funding the dealer, so the dealer benefits from not paying the full funding cost. This looks structurally similar to DVA.

Scenario	DVA Treatment	FVA Treatment	Resolution
Dealer in-the-money (asset)	No DVA impact	FCA: dealer pays funding cost	No overlap
Dealer out-of-the-money (liab)	DVA gain: own-default benefit	FBA: funding benefit	Overlap risk — decompose carefully
Full daily-margined CSA	DVA reduced	FVA ≈ 0 (collateralised)	No issue
Asymmetric CSA	Partial DVA	Partial FVA/FBA	Careful scenario decomposition needed

The Burgard-Kjaer Resolution

Burgard and Kjaer (2013) showed that the correct approach is a unified semi-replication framework that incorporates the dealer's own default explicitly. In this framework, DVA and FBA are mutually exclusive by construction: FBA applies only when the dealer is alive and funding trades; DVA applies when the dealer defaults. The unified valuation equation is:

$$V_{total} = V^{RO} + CVA + DVA + FCA + FBA^* \cdot (1 - R^D)$$

V^{RO} = risk-free value; CVA = counterparty credit adjustment (negative); DVA = dealer own-default benefit (positive); FCA = funding cost adjustment (negative); FBA = funding benefit adjustment adjusted for dealer recovery rate R^D to avoid double-count with DVA*

In practice, most dealers resolve this pragmatically: DVA is recognised for accounting purposes (IFRS 13 requires fair value to reflect own credit risk), while FVA is charged by treasury as an internal funding transfer price. The risk of double-counting is managed through careful definition of which scenarios each adjustment covers.

7. Worked Numerical Example: FVA on an Uncollateralised IRS

Trade Setup

EUR 100M notional, 5-year plain vanilla interest rate swap. Dealer pays fixed 3.50% annually and receives 6-month EURIBOR. No CSA in place (uncollateralised). Dealer internal funding spread: 80 bps above OIS.

Parameter	Value	Source / Assumption
Notional	EUR 100,000,000	Trade terms
Maturity	5 years	Trade terms
Fixed rate (dealer pays)	3.50% p.a.	At-the-money at inception
Floating rate (dealer receives)	6M EURIBOR	Market rate
OIS rate (flat)	3.20%	Market data
Dealer funding spread s^O	80 bps (0.80%)	Internal treasury curve
All-in funding rate	4.00%	OIS + spread
Rate volatility σ^R	80 bps normal vol	Market implied

Step 1: EPE and ENE Profile

Using a simplified closed-form approximation for an at-the-money IRS, the **Expected Positive Exposure (EPE)** at time *t* for a receiver of floating is:

$$EPE(t) \approx N \cdot \sigma^R \cdot \sqrt{[t \cdot (T-t) / T]}$$

N = notional; σ^R = annualised normal rate volatility (80 bps); T = maturity (5 years); t = evaluation time. ENE(t) = −EPE(t) by symmetry for at-the-money swap.

Time t (yrs)	EPE(t) EUR M	ENE(t) EUR M	B(0,t)	FCA contrib (EUR)	FBA contrib (EUR)
0.5	1.26	−1.26	0.9841	−49,582	+49,582
1.0	1.77	−1.77	0.9685	−137,227	+137,227
1.5	2.13	−2.13	0.9531	−162,255	+162,255
2.0	2.40	−2.40	0.9380	−180,096	+180,096
2.5	2.60	−2.60	0.9232	−191,922	+191,922
3.0	2.71	−2.71	0.9085	−196,856	+196,856
3.5	2.71	−2.71	0.8942	−193,813	+193,813
4.0	2.56	−2.56	0.8800	−179,917	+179,917
4.5	2.16	−2.16	0.8661	−149,866	+149,866
5.0	1.41	−1.41	0.8524	−96,119	+96,119

Time t (yrs)	EPE(t) EUR M	ENE(t) EUR M	B(0,t)	FCA contrib (EUR)	FBA contrib (EUR)
TOTAL				**−EUR 1,537,653**	**+EUR 1,537,653**

FCA/FBA = s^0 × EPE/ENE(t) × B(0,t) × Δt (Δt = 0.5). Each row is semi-annual.

Step 2: Net FVA and Key Insight

```
FVA (net) = FCA + FBA = −EUR 1,537,653 + EUR 1,537,653 = EUR 0
```

At-the-money at inception: EPE = |ENE| ⇒ FCA = FBA exactly ⇒ Net FVA = 0. After a 50 bp adverse rate move (trade moves €2.4M ITM): Net FVA ≈ −EUR 480,000.

This illustrates the critical insight: FVA is zero at inception for at-the-money trades because positive and negative exposure cancel. FVA becomes material as the trade moves in- or out-of-the-money. The 80 bps funding spread on €2.4M positive exposure over the remaining life generates approximately €480,000 in FCA — a hidden cost that must be funded from treasury.

8. MVA: Formal Framework & Formula Derivation

8.1 Origins: The UMR Revolution

MVA did not exist as a pricing consideration before 2015. It was created by regulation: the Uncleared Margin Rules (UMR), agreed by BCBS-IOSCO in 2015, required bilateral (non-cleared) derivatives above certain thresholds to post Initial Margin (IM) — a buffer against future potential exposure. Critically, this IM must be segregated (held by a third-party custodian) and cannot be rehypothecated. The capital is genuinely locked up.

The phase-in ran from September 2016 (largest dealers) to September 2022 (Phase 6, covering buy-side firms with average aggregate notional above USD 8 billion). By 2023, an estimated USD 1.3 trillion in IM was being posted globally under bilateral UMR.

8.2 The ISDA SIMM Model

The industry standard for computing bilateral IM is the ISDA Standard Initial Margin Model (SIMM), introduced in 2016. SIMM is a sensitivity-based model: it takes the first-order sensitivities (delta, vega) of the portfolio to each risk factor and applies regulatory-calibrated risk weights and a correlation structure.

$$IM_{class} = \sqrt{\left(\Sigma^I \, \Sigma_j \, \rho^I{}_j \cdot WS^I \cdot WS_j \right)}$$

WS^I = weighted sensitivity = $\delta^I \cdot RW^I$ (delta times SIMM risk weight for bucket i); ρ^I_j = SIMM inter-bucket correlation; aggregated across risk classes (IR, Credit, Equity, FX, Commodity)

8.3 MVA Formula and Intuition

MVA is the present value of the expected opportunity cost of posting IM over the life of the trade. Let *IM(t)* denote the expected IM requirement at time *t* (computed under SIMM), and let $s_m(t)$ denote the spread between the cost of funding IM and the rate earned on posted IM (typically OIS):

$$MVA = -\int_0^T s_m(t) \cdot E[IM(t)] \cdot B(0,t) \, dt$$

$s_m(t)$ = dealer funding rate − IM remuneration rate (e.g., OIS or Fed Funds); E[IM(t)] = expected SIMM IM at time t (path-dependent, requires simulation); B(0,t) = OIS discount factor; MVA is always negative (cost to poster)

Discrete (simulation) form:

$$MVA \approx -\Sigma^I \, s_m(t^I) \cdot E[IM(t^I)] \cdot B(0,t^I) \cdot \Delta t$$

$E[IM(t^I)]$ = cross-sectional average of SIMM IM across Monte Carlo paths at time step i; Δt = simulation time step; sign convention: MVA < 0 always

Feature	Detail
IM computation	ISDA SIMM (bilateral) or CCP SPAN/TIMS (cleared)
Remuneration rate	OIS/ESTR typically; some CSAs specify Fed Funds
Funding spread s_m	Typically 20–100 bps; varies by bank rating and market conditions

Feature	Detail
IM profile shape	Highest at inception (high uncertainty); declines as maturity approaches
MVA sign	Always negative — posting IM is always a cost to the poster
Key driver post-2022	Phase 6 UMR bringing mid-size buy-side firms into scope

9. Worked Numerical Example: MVA Under ISDA SIMM

Trade Setup (same EUR 100M 5Y IRS as Section 7)

The trade is now subject to bilateral UMR (Phase 5/6). Initial Margin is computed under ISDA SIMM. Dealer funding cost: 4.00% p.a. IM remuneration rate: 3.20% (OIS). Funding spread s_m = 80 bps.

Step 1: SIMM IM at Inception

For an IRS under ISDA SIMM v2.6, IM is driven by the IR delta risk class. The portfolio DV01 for a EUR 100M 5Y IRS is approximately EUR 45,000 per bp. Scaled to SIMM sensitivity units (δ = DV01 × 10,000 / notional × notional = EUR 4,500,000 in sensitivity units). SIMM Risk Weight for EUR 5Y bucket: RW = 75 bps (v2.6 calibration).

$$\text{SIMM IM} \approx |\delta| \cdot RW^{IR} = \text{EUR } 4{,}500{,}000 \times 75 \text{ bps} = \text{EUR } 3{,}375{,}000$$

Simplified single-bucket approximation. Actual SIMM aggregates across tenor buckets with correlation matrix ρ^I_j, producing a lower figure due to netting. Single-bucket provides conservative upper bound.

Step 2: IM Amortisation Profile & MVA Calculation

IM declines as the trade ages and residual maturity shortens. We model linear amortisation from EUR 3.375M at inception to zero at maturity:

Time t (yrs)	E[IM(t)] EUR M	s_m (bps)	B(0,t)	MVA contrib (EUR)
0.5	3.038	80	0.9841	−11,960
1.0	2.700	80	0.9685	−20,877
1.5	2.363	80	0.9531	−18,014
2.0	2.025	80	0.9380	−15,194
2.5	1.688	80	0.9232	−12,451
3.0	1.350	80	0.9085	−9,812
3.5	1.013	80	0.8942	−7,253
4.0	0.675	80	0.8800	−4,752
4.5	0.338	80	0.8661	−2,347
5.0	0.000	80	0.8524	0
TOTAL MVA				**≈ −EUR 102,660**

MVA contrib = s_m × E[IM(t)] × B(0,t) × Δt. Linear IM amortisation is illustrative; actual SIMM profiles are non-linear.

Step 3: Total XVA for This Trade

XVA Component	Amount (EUR)	Notes
CVA (Module 3)	−€85,000	Counterparty BBB, 5Y CDS ≈ 100 bps

XVA Component	Amount (EUR)	Notes
DVA (Module 3)	+€32,000	Dealer A-rated, 5Y CDS ≈ 50 bps
FVA (net, at-the-money)	≈ EUR 0	Symmetric at inception; non-zero after rate moves
MVA	−€102,660	UMR bilateral IM, s_m = 80 bps, SIMM v2.6
Total XVA Charge	≈ −EUR 155,660	Added to bid-offer or charged explicitly

This illustrates why XVA desks exist: a trade that appears flat or slightly profitable on a risk-neutral basis carries approximately EUR 156K in all-in adjustments. Ignoring these costs means the trading desk books profits that the bank simultaneously pays out of treasury and regulatory capital.

10. Regulatory Treatment: FRTB-CVA, SA-CCR & UMR Interactions

10.1 FVA and Regulatory Capital

FVA is a P&L adjustment. It interacts with capital in several important ways:

- **IFRS 13 / GAAP:** 'Exit price' fair value effectively requires FVA inclusion for uncollateralised derivative positions.
- **Deferred Tax Assets:** FVA reserves create DTA/DTL positions that affect CET1 capital under Basel III deduction rules.
- **Hedging of FVA:** Banks attempt to hedge FVA risk using CDS or repo instruments. Hedge positions themselves consume market risk capital under FRTB.

10.2 MVA and the UMR Regulatory Framework

Regulation	Jurisdiction	MVA Impact
BCBS-IOSCO UMR (2015)	Global	Creates IM requirement; primary driver of MVA
EMIR Refit	EU	Phase 5/6 brings pension funds into scope; increases MVA
Dodd-Frank / SEC Rule (2022)	US	Extended scope to equity derivatives; new MVA populations
SA-CCR (Basel III)	Global	Higher EAD increases cleared IM requirements
FRTB-CVA Framework	Global	FVA hedges NOT recognised as CVA hedges → regulatory/economic basis
CRR III (EU, 2025)	EU	Output floor may increase RWA; indirectly raises funding costs

10.3 FRTB-CVA Interaction

Under FRTB-CVA (SA-CVA), banks compute sensitivity-based capital charges covering the market risk of CVA itself. FVA hedges — instruments used to manage FVA risk — are not recognised as CVA hedges under FRTB-CVA, creating a basis: the bank may have economic hedges that do not reduce regulatory capital. This is a material ongoing industry concern.

10.4 The Cleared vs Bilateral Choice

Factor	Bilateral (OTC)	Cleared (CCP)
CVA/DVA	Applies	Zero (CCP-guaranteed)
FVA	Applies if no or partial CSA	Near-zero (daily VM settlement)
MVA	High (SIMM IM, segregated, no rehypo)	Lower (CCP model, portfolio netting)
VM settlement	Periodic (may not be daily)	Daily (CCP requirement)

Factor	Bilateral (OTC)	Cleared (CCP)
Netting benefit	Within bilateral netting set only	Across CCP member portfolio

The choice to clear is now a quantitative optimisation: total XVA (CVA + FVA + MVA + KVA) must be computed for both routes, along with operational costs and netting benefits.

11. Crisis Case Studies

11.1 Lehman Brothers Collapse (September 2008)

When Lehman filed for Chapter 11 on 15 September 2008, it had an OTC derivatives portfolio of approximately 900,000 contracts with notional value exceeding USD 35 trillion. The resolution exposed several FVA-relevant failures:

- **Unsecured exposure:** Many trades lacked robust CSAs. Counterparties discovered that receivables they expected to collect were now unsecured claims against the bankruptcy estate.
- **Funding spiral:** As Lehman's credit deteriorated in the months before default, its funding costs surged. Internal funding spreads reportedly exceeded 300 bps in August 2008 — a classic FCA squeeze.
- **Repo market freeze:** Lehman relied heavily on repo for short-term funding of its derivatives portfolio. When repo counterparties withdrew facilities, Lehman faced immediate liquidity failure.
- **Resolution cost:** Winding down the portfolio through the PwC administration took years and cost billions. Had FVA been properly charged, Lehman's pricing would have reflected the true cost of its high funding spread.

11.2 The 2008–2009 Funding Crisis: Systemic FVA Implications

The broader 2008 crisis created the conditions that made FVA a mainstream concern. The LIBOR-OIS spread — a key measure of bank funding stress — widened from approximately 10 bps pre-crisis to over 360 bps in October 2008. This meant the difference between risk-neutral (OIS) pricing and real-world (bank funding) pricing had grown by more than 35x. Banks that had been ignoring this spread suddenly faced P&L holes attributable not to trading losses but to funding losses that should have been reserved from day one.

11.3 JP Morgan FVA Charge (January 2014)

JP Morgan's Q4 2013 earnings call included the first public quantification of FVA: a USD 1.5 billion loss booked against Q4 2013 results. Goldman Sachs, Bank of America, and Morgan Stanley followed within months, collectively booking multi-billion dollar FVA adjustments. The market interpreted this as validation that the entire industry had been systematically underpricing uncollateralised derivatives for years.

11.4 UMR Phase 6 (September 2022): The MVA Inflection Point

Phase 6 of Uncleared Margin Rules brought mid-size asset managers, hedge funds, pension funds, and corporates with average aggregate notional above USD 8 billion into scope. Global bilateral IM under UMR reached USD 1.3 trillion by end-2022, up from USD 130 billion before Phase 1 (2016). Key consequences:

- **IM amounts:** USD 1.3 trillion in bilateral IM posted globally by end-2022.
- **MVA impact:** For a mid-size pension fund with EUR 5B in bilateral derivatives, the annual MVA cost increased by an estimated EUR 15–40M.

- **Clearing incentive:** Many buy-side firms moved bilateral trades to CCPs to reduce SIMM IM (which is higher than CCP IM for diversified portfolios due to model conservatism).
- **Trade compression:** MVA incentivised systematic portfolio compression — reducing gross notional to lower IM requirements without changing net risk.

12. Advanced Topics

12.1 FVA Asymmetry: Wrong-Way Risk

Standard FVA computation assumes the funding spread $s^o(t)$ is independent of the exposure profile V(t). In reality, there can be significant correlation — the funding spread may widen precisely when the exposure also widens:

- **Wrong-way FVA:** If the bank's credit worsens (funding spread widens) at the same time the trade moves against it (exposure increases), the true FVA is higher than the independent calculation suggests.
- **Right-way FVA:** If the bank's credit improves when exposure increases, FVA may be lower than the base case.
- **Modelling approach:** Capturing this correlation requires joint simulation of market risk factors and the bank's own credit spread — computationally intensive but increasingly required for accurate XVA.

12.2 Bilateral vs Unilateral FVA

Framework	Description	Implication
Unilateral FVA	Only the dealer's funding costs considered	Asymmetric: counterparty has different FVA. Price depends on who calculates.
Bilateral FVA	Both parties' funding costs averaged	Symmetric: both parties agree on price. Requires counterparty funding info.
Semi-replication (Burgard-Kjaer)	Funding costs embedded in replication strategy with own default event explicitly modelled	Theoretically consistent; avoids DVA/FVA double-count by construction.

12.3 ColVA — Preview of Module 6

Collateral Valuation Adjustment (ColVA) arises from the optionality embedded in collateral agreements. A CSA typically allows the collateral poster to deliver any eligible security from an approved schedule. The poster will rationally deliver the cheapest-to-deliver asset. The value of this optionality — the difference between the ideal collateral and the cheapest available — is ColVA. Module 6 will cover ISDA netting agreements, CSA architecture, and ColVA in full.

12.4 The XVA Optimisation Problem

For a sophisticated dealer, the question is not just "what is FVA?" but "how do we minimise total XVA across the book?" This optimisation involves:

- **Trade compression:** Terminating offsetting bilateral trades to reduce gross notional and IM requirements.
- **Netting set optimisation:** Routing new trades through the netting set where they provide the greatest EPE reduction.

- **Clearing decisions:** Choosing whether to clear based on total XVA (CVA + FVA + MVA + KVA) rather than individual components.
- **CSA renegotiation:** Moving from zero-threshold CSAs to daily VM posting to reduce FVA (at the cost of operational burden).
- **Collateral transformation:** Using securities lending or repo to convert ineligible assets into eligible IM, reducing the opportunity cost captured in MVA.

13. References

Regulatory & Official Documents

[1] BCBS-IOSCO (2015). Margin requirements for non-centrally cleared derivatives. Bank for International Settlements.

[2] European Market Infrastructure Regulation (EMIR) — Regulation (EU) No 648/2012 and Refit amendments (2019).

[3] ISDA (2023). ISDA SIMM Methodology, version 2.6. International Swaps and Derivatives Association.

[4] BCBS (2020). Minimum capital requirements for market risk (FRTB). Bank for International Settlements.

[5] SEC (2022). Security-Based Swap Margin and Segregation Requirements — Final Rule.

[6] European Commission (2023). Capital Requirements Regulation III (CRR III) — Basel IV implementation in the EU.

Academic Papers

[7] Piterbarg, V. (2010). Funding beyond discounting: Collateral agreements and derivatives pricing. Risk, February 2010.

[8] Burgard, C. and Kjaer, M. (2011). Partial differential equation representations of derivatives with bilateral counterparty risk and funding costs. Journal of Credit Risk, 7(3).

[9] Hull, J. and White, A. (2012). The FVA debate. Risk, July 2012.

[10] Burgard, C. and Kjaer, M. (2013). Funding costs, funding strategies. Risk, December 2013.

[11] Andersen, L., Duffie, D. and Song, Y. (2019). Funding value adjustments. Journal of Finance, 74(1), 145–192.

[12] Green, A. (2015). XVA: Credit, Funding and Capital Valuation Adjustments. Wiley Finance.

[13] Albanese, C. and Andersen, L. (2014). Accounting for OTC derivatives: Funding adjustments and the re-hypothecation option. SSRN Working Paper.

Accounting & Valuation Standards

[14] IFRS 13 — Fair Value Measurement. International Accounting Standards Board, 2011.

[15] ASC 820 — Fair Value Measurement. Financial Accounting Standards Board (US GAAP).

Market & Industry Sources

[16] JP Morgan Chase (2014). Q4 2013 Earnings Release, January 14, 2014 — FVA charge disclosure.

[17] ISDA (2023). Global Derivatives Statistics — Margin Survey 2023.

[18] Deloitte (2022). Uncleared Margin Rules Phase 6: Readiness and Implementation Guide.

14. Credits & Methodology Note

Author

Sarthaka Nanda

AI Collaboration

This module was produced in collaboration with Anthropic Claude as part of the Practitioner Knowledge Repository on Banking Regulation and Risk Management. The collaboration follows the 20-60-20 principle:

- **20% Human prompting & instruction:** Topic selection, structural design, strategic framing, worked example parameters, and quality direction.
- **60% AI generation:** Technical content drafting, formula derivation, table construction, regulatory mapping, and case study research.
- **20% Human review & editing:** Mandatory review for accuracy, consistency, professional standards, and editorial judgment before publication.

AI-generated content without mandatory human review is not professional-grade content — it is a liability.

Series

Module 1: Basel Accord Evolution | Module 2: FRTB | Module 3: CVA, DVA & CCR | Module 4: FVA & MVA | Module 5: KVA & Capital Costs (forthcoming) | Module 6: ISDA, Netting & ColVA (forthcoming)

Licence

Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). Free to share and adapt for non-commercial purposes with attribution.